

Demonstrating WAVY: a hand wearable with force and vibrotactile feedback for multimodal interaction in Virtual Reality

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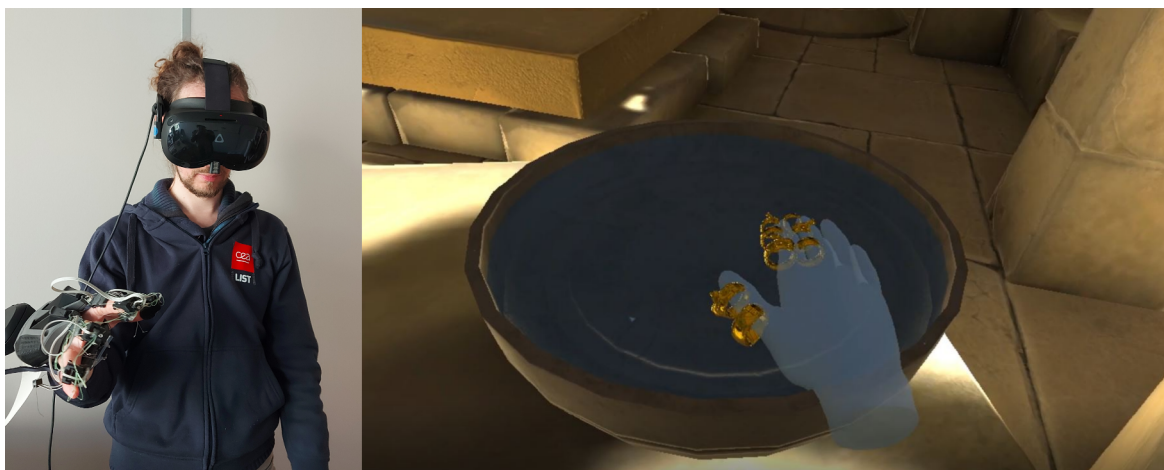


Figure 1: WAVY setup and VR scene

ABSTRACT

While the public has access to audio and visual stimuli in virtual reality (VR) thanks to affordable and light headsets, the haptic sense (i.e. the sense of touch) is often lacking or restricted to controllers' vibrations. Besides, commercial haptic wearables remain too expensive for the mass market. Thus, we designed an affordable and light hand wearable that combines tracking, multi-point vibrations and force feedback for the palm and the fingers. The wearable is demonstrated in a VR scenario.

Index Terms: Wearable, Haptic, Vibrotactile Feedback, Force Feedback, Multimodal Interaction

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1 INTRODUCTION

Haptic cues such as vibrations or force are well known to help complete difficult tasks such as manipulation [5] or learning technical gestures [2, 4, 6]. Thus, to improve immersion in virtual reality, it is desirable to have access to such feedback.

This has led to a wide development of wearables and especially hand wearables [5] (i.e. interfaces that are worn on the user's body) as they can enable the user to move freely in a large area while continuing to perceive the haptic feedback.

However, these devices generally only focus on one type of haptic feedback (i.e. force or vibration) and more often on vibration than force feedback [5]. Moreover, other wearable providing both feedback often do not provide a large workspace area due to the need to remain tethered to their power supply (e.g. compressor like on Haptx's devices [1] or plug like on the CyberGrasp device [8]).

Thus, we propose the design of a fully embedded, wearable for virtual reality that enables several types of customizable force feedback and vibration on a large range of the haptic bandwidth. To achieve this goal, we have implemented different types of technologies and specifically developed a novel type of safe, lightweight and low-cost actuator that is able to generate both force and vibration feedback separately or simultaneously [7]. Moreover, our device is able to stream haptic signals from Unity in real time.

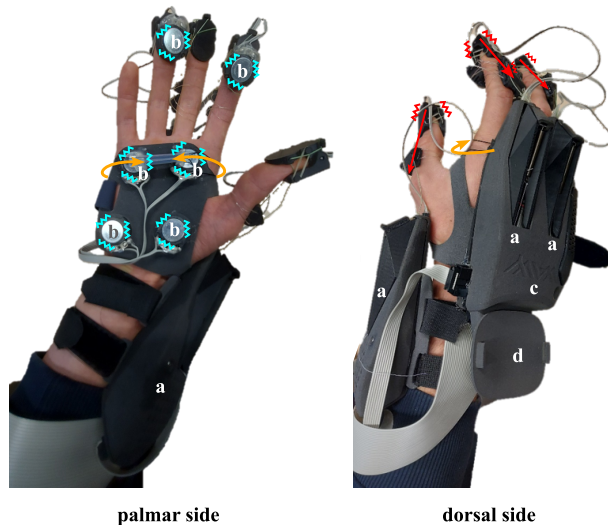


Figure 2: Wearable: Haptic rendering module with a: WAL module producing resistive force and vibration in red, b: voice coil actuators producing vibrations in cyan, c: motor position (hidden below the case) producing force in orange, d: Hand Tracker location

2 DEMONSTRATOR DESCRIPTION

The wearable was customized to fit the right hand and allows force and vibration feedback on several parts on it. The device includes a haptic rendering part and a control part. The haptic rendering part is worn on the hand while the control part can be worn in a neckband.

The haptic rendering part is illustrated in Fig. 2 and weighs 330 g. It is composed of three actuation technologies. First, three modules of WAL (Wearable based on Active Lubrication) [7] (see a in Fig. 2) use our novel active lubrication brakes. These enable customizable force and vibration feedback on the thumb, index and middle finger when these fingers execute a closing movement (see red marking in Fig. 2). The actuators enable force feedback between 0.45 and 3.15 N and vibration feedback from 1 Hz to 100 Hz up to 0.2 N of amplitude. This technology allows the creation of interactions with rigid objects or deformable ones as well as textures. Second, six voice coil actuators E-12041808 (Soberton Inc). (see b in Fig. 2) enable vibrations (see cyan marking on Fig. 2) on the palm (4 positions) and fingertip (index and ring finger) on a large range of the haptic bandwidth (5-1000 Hz). These actuators simulate various light touches and exploit the apparent motion illusion for rendering continuous interactions [3]. Third, one electromagnetic motor MG16B-0306AB-00 (Copal Electronics) (see c in Fig. 2) provides squeezing feedback on the side of the palm (up to 20 N applied to the cable) as well as other sensations while holding objects (e.g. a beating heart).

The control part is composed of two electronic boards that control the device (one for the vibration and motor and one for the force feedback on the fingers i.e. WAL modules). They are powered from two 3-cells 2200mAh li-ion battery and communicate with the computer from a USB cable or by WIFI. All the vibration and force data are streamed in real time from Unity. The sample rate of the haptic feedback is set at 4 kHz and the communication rate of the data buffer at 500 Hz. We developed a Unity package that includes the hardware communication and control module, the tools to design and manage the haptic signals and some interaction example sets.

To immerse the users in the virtual environment, we use a Vive Focus 3 headset as PCVR devices with the OpenXR Unity Plugin. We use the embedded Vive Focus 3 Hand tracking system for the

left hand (not equipped hand). For the Right hand, we use a VIVE Ultimate Tracker (see d in Fig. 2) to track the global hand position and the WAL module to track the fingers positions (response time up to 2.2 ms).

To illustrate the type of haptic feedback enabled by this device, we designed one short scenario in virtual reality where the participant is invited to interact with several virtual objects relying on different haptic interactions.

3 CONCLUSION

This paper presents a novel haptic wearable able to render both force and vibration feedback on the palm and fingers of the hand and able to play them in real-time virtual reality scenes built with the Unity engine.

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